Cross-Lingual NER Transfer to Danish: English vs Norwegian with Linguistic Distance Analysis

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Abstract

We study cross-lingual named entity recognition (NER) transfer to Danish using two multilingual transformer encoders — mBERT and XLM-R — pitting English against Norwegian as source languages across three random seeds. On the held-out Danish test set, Norwegian comes out ahead in the zero-shot setting: +1.86 F1 for XLM-R ($p = 0.10$) and +3.36 F1 for mBERT ($p = 0.046$, significant at $\alpha = 0.05$). One run — seed 42 — looked especially favorable to Norwegian XLM-R on the development set, then revealed itself as an outlier once averaged across seeds, which is precisely why cross-lingual source-language comparisons should not rest on a single initialization. The results track what typological structure would lead you to expect: Norwegian is roughly 5.7× closer to Danish than English in WALS syntactic-feature space (cosine distance 0.021 vs. 0.119). The picture changes once Danish fine-tuning enters: an English mBERT checkpoint achieves the best transfer-plus-fine-tune dev score (89.4 strict F1), suggesting that a large source corpus can offset typological distance once target-language supervision is available. We additionally report unlabeled and loose span F1, per-entity-type breakdowns, a Danish data-efficiency curve, and a diagnosis of one misaligned prediction file.

1 Introduction and Background

Named entity recognition (NER) is the task of identifying text spans that refer to persons, locations, organizations, and related categories. For Danish, the bottleneck is typically data rather than model capacity. English has abundant annotated resources; Danish does not. Cross-lingual transfer tries to work around this constraint by training on a well-resourced language and deploying to the target with no labels — or before target labels exist.

The real question is whether typological closeness actually predicts transfer quality. Do Norwegian-trained models transfer to Danish better than English-trained ones? Norwegian is an obvious candidate: it shares substantial syntax and vocabulary with Danish, both being North Germanic languages. English is the conventional high-resource default. Keeping the encoder family fixed across source languages — mBERT and XLM-R in both cases — means differences can be attributed to the source language, not to swapping architectures.

We run five experiment families: monolingual source baselines, zero-shot transfer to Danish, transfer followed by Danish fine-tuning, Danish-only models at varying annotation budgets, and a typological-distance analysis. Distances are computed from WALS feature vectors via `lang2vec` (Littell et al., 2017), with Swedish and German included as reference points beyond the core English–Norwegian contrast. Statistical reliability is assessed with paired resampling tests.

The report proceeds through related work, research questions, dataset description, typological analysis, methodology, implementation notes, results, interpretation, and concluding remarks.

2 Project Goal

The primary goal is straightforward to state but less straightforward to answer cleanly: is Norwegian a more effective transfer source for Danish NER than English? The hypothesis behind it is reasonable — languages sharing closer history and structure should induce more compatible representations inside a multilingual encoder, which should translate into stronger cross-lingual transfer. Rather than treating this as self-evident, this study operationalizes the intuition as a controlled empirical comparison.

The practical angle matters as much as the theoretical one. Raw F1 is useful, but the real decision facing a practitioner is: when Danish annotation is scarce, which source language and encoder com-

Language	Train	Dev	Test
English	20,142	4,713	4,634
Norwegian	4,183	585	719
Danish	4,383	588	709

Table 1: Universal NER sentence counts per split. English is substantially larger, making it the natural high-resource source baseline.

bination is the best bet? A consistent Norwegian advantage makes the choice obvious. A negligible difference shifts the decision toward corpus size and encoder selection instead.

A secondary aim is to untangle three influences that tend to get conflated: source-language choice, encoder architecture, and the amount of available Danish supervision. Placing monolingual baselines, zero-shot transfer, fine-tuned transfer, and Danish-only models side by side in a single study makes it possible to assign credit to each factor rather than leaving them entangled.

3 Dataset

We use the Universal NER benchmark (Mayhew et al., 2024) for English, Norwegian and Danish. Data are provided sentence-by-sentence in token-aligned IOB2 format (B-TYPE, I-TYPE, O), with three entity types in the primary label column: PER (person), LOC (location) and ORG (organization).

One detail warrants attention upfront. The Danish file does contain MISC labels, but they reside in a *secondary* annotation column — not the primary column that the training and evaluation scripts read. As a result, MISC F1 is effectively forced to 0.0 for every Danish-target system: not because the model cannot learn MISC, but because the pipeline never sees those labels in the first place (Section 10 provides details).

The three languages play distinct roles. English is the high-resource source; Norwegian is the closely related North Germanic alternative; Danish is the low-resource target whose labels we either restrict or withhold in the transfer scenarios.

The size asymmetry is lopsided in an important way: English provides roughly five times as many training sentences as Danish or Norwegian (Table 1). That data-volume advantage for the English source is a separate factor from typological proximity, and the fine-tuning experiments are partly about disentangling the two. Danish and Norwegian are closely matched in size, which simplifies interpretation of the Danish data-budget conditions (10%, 25%, 50%).

Because the primary annotation column uses the same label inventory across all three languages, observed performance differences are attributable to language transfer rather than annotation-schema mismatches. We verified this consistency before training.

4 Related Work

Cross-lingual NER is a crowded area, especially for low-resource targets. English-to-Danish transfer has already been shown to work as a practical zero-shot baseline, improving quickly once even modest Danish supervision is added (Plank, 2020). Our results broadly corroborate that picture while adding a direct Norwegian comparison. More broadly, work such as Fu et al. (2023) finds that structural similarity often predicts transfer success — a claim we pressure-test here with a controlled setup that holds architecture, data format, and evaluation protocol constant.

Two Danish-specific resources anchor the fully supervised baseline literature. DaNE (Hvingelby et al., 2020) provides the manually annotated corpus that underlies most subsequent Danish NER work, and DaCy (Enevoldsen et al., 2021) demonstrates strong Danish-supervised pipelines built on the same annotation conventions. Our Danish monolingual baselines are most directly comparable to that line of work.

Multilingual encoders make zero-shot transfer practically viable. BERT established the transformer pretraining paradigm (Devlin et al., 2019), and XLM-R extended it with a substantially larger and more linguistically diverse corpus (Conneau et al., 2020), often yielding better cross-lingual generalization. Whether that edge persists when source and target are as close as Norwegian and Danish is an open question that motivates our design.

Typological distance as a predictor of transfer performance has been explored via resources such as URIEL+ (Khan et al., 2025) and DistALS (Van Der Goot et al., 2025); recent studies argue that surface similarity can explain a substantial portion of the zero-shot gap (Sohn et al., 2024). Our contribution is to compute WALS-feature distances explicitly and tie those measurements directly to observed F1 differences, using the Universal NER benchmark (Mayhew et al., 2024) across all three languages to keep annotation-format differences out of the picture.

5 Research Questions

Three questions drive the study.

1. **How do monolingual baselines compare to cross-lingual transfer routes?** We look at English and Norwegian source baselines, zero-shot transfer to Danish, and Danish-only supervision.

2. **What is lost in zero-shot transfer, and how much does Danish fine-tuning recover?** We measure the extent to which modest Danish annotation closes the gap to fully supervised Danish training.

3. **Does lower typological distance to Danish predict better transfer?** We compare English-source and Norwegian-source transfer against quantified typological distances to test this hypothesis.

The third question is the real centerpiece of the study. Showing that cross-lingual transfer works is a necessary starting point; understanding whether structural proximity is the active ingredient requires holding architecture, data format, and evaluation protocol constant while varying only the source language.

6 Methodology

6.1 Experimental design

We run five families of experiments:

- monolingual baselines for English, Norwegian and Danish,
- zero-shot transfer to Danish from English and Norwegian,
- transfer fine-tuning, where a source-trained model is further trained on Danish,
- Danish-only supervision at multiple data budgets (10%, 25%, 50%, 100%), and
- linguistic-distance analysis based on WALS typological feature vectors.

The label space and Danish target splits stay fixed throughout. What varies are source language, encoder, and Danish supervision budget — three clean axes that keep the interpretation tractable.

Hyperparameter	Value
Learning rate	2×10^{-5} (AdamW)
Train batch size	8
Eval batch size	16
Max sequence length	256 tokens
Training epochs	4
Weight decay	0.01
Random seed	42
Early stopping	Best checkpoint by dev F1

Table 2: Shared hyperparameters used across all experiments. Multi-seed replication and its implications are discussed in Section 11.

6.2 Models and checkpoints

Two multilingual encoders serve as the backbone for token-classification:

- **mBERT**: `bert-base-multilingual-cased` (Devlin et al., 2019) — 12 layers, 110M parameters, trained on 104 languages.
- **XLM-R**: `xlm-roberta-base` (Conneau et al., 2020) — 12 layers, 125M parameters, trained on 100 languages with a substantially larger pretraining corpus than mBERT.

An English-only BERT baseline (`bert-base-cased`) provides a single-language reference point. All other experiments attach a linear token-classification head for BIO tagging on top of the multilingual encoders.

For fine-tuned transfer (EN→DA FT, NO→DA FT) we initialize from the best zero-shot checkpoint and continue training on Danish. These models have therefore been exposed to labeled data from both the source and the target language.

6.3 Training hyperparameters

All runs share the same configuration unless otherwise stated:

We deliberately avoided per-language hyperparameter tuning, applying the same configuration to every run. That choice can leave performance on the table for individual systems, but it keeps comparisons interpretable.

6.4 Evaluation

The primary metric is strict span-level F1 as computed by `span_f1.py`, requiring an exact match on both entity boundaries and type. We also report:

- unlabeled span F1 (exact boundaries, any label),

- loose span F1 (partial overlap, matching label).

These additional views help separate boundary errors from label confusions from entirely missed spans, giving a more complete picture of where each system struggles.

Because the English test split is masked (no gold labels), test-set comparisons focus on Danish and Norwegian. Danish-target models are evaluated on the Danish DDT test split; Norwegian source baselines on the NorNE test split.

6.5 Statistical testing

To assess the English–Norwegian contrast we apply paired resampling tests (paired bootstrap, 10,000 iterations) to dev and test prediction files. With only a few seeds, these tests serve as a practical reliability check, not a substitute for broader replication. More seeds would still be preferable.

7 Code and Implementation

The core training pipeline lives in `Baselines/train_ner_baseline.py`. Launcher scripts in `Baselines/` select language, model and output directory, which keeps runs comparable and reduces the risk of configuration drift between experiments.

The shared script executes the following steps:

1. read IOB2 data from disk,
2. construct the label vocabulary,
3. tokenize sentences with the chosen multilingual transformer tokenizer,
4. align word-level labels to subword tokens,
5. fine-tune the token-classification model,
6. evaluate on the development set, and
7. write predictions and metrics to the output directory.

Label alignment deserves a brief note. Individual words can decompose into multiple subword tokens, so the implementation assigns the label to the first subword and masks subsequent ones in the loss — a common convention that keeps training stable and ensures that evaluation operates on token-level outputs consistent with the gold annotations.

The pipeline emits `dev_metrics.json`, `dev_predictions.iob2` and `test_predictions.iob2` for every run. Having those artifacts made it straightforward to spot anomalies: sentence-count mismatches between prediction files and gold splits are immediately visible when you check them systematically (see Section 10).

The typological-distance computations are handled by a separate utility that operates on `lang2vec` WALS vectors. It is fully decoupled from model training and exists only to supply the structural evidence used to interpret observed transfer gaps.

8 Linguistic Distance Analysis

8.1 Motivation

If Norwegian is structurally closer to Danish in syntax and lexicon, it may offer a stronger transfer signal. The hypothesis sounds reasonable. The subsections below put a number on that closeness, then compare it to the observed F1 differences to see how well the two align.

8.2 Method

We compute typological feature vectors with `lang2vec` (Littell et al., 2017), using the `syntax_wals` feature set from WALS. This representation encodes approximately 103 syntactic and morphological features; features unavailable for a given language pair are excluded before computing cosine distance. Formally,

$$d(L_1, L_2) = 1 - \frac{\mathbf{v}_1 \cdot \mathbf{v}_2}{\|\mathbf{v}_1\| \, \|\mathbf{v}_2\|},$$

where each vector $\mathbf{v}_i$ contains only the jointly available features for that pair.

8.3 Results

Table 3 lists pairwise cosine distances for English, Danish, Norwegian (Bokmål), Swedish and German. Swedish and German serve as reference points: Swedish sits close to Danish within the North Germanic cluster, while German occupies a more distant West Germanic position.

The ordering is clear:

- Norwegian is the closest language to Danish in this space (0.021) — roughly 5.7× closer than English (0.119).

	English	Danish	Norwegian	Swedish	German
English	0.000	0.119	0.109	0.080	0.145
Danish	0.119	0.000	**0.021**	0.039	0.124
Norwegian	0.109	0.021	0.000	0.065	0.102
Swedish	0.080	0.039	0.065	0.000	0.164
German	0.145	0.124	0.102	0.164	0.000

Table 3: Pairwise cosine distances based on WALS syntactic features (lang2vec `syntax_wals`). Lower = more similar. Norwegian is the closest language to Danish (0.021), followed by Swedish (0.039); English and German are substantially more distant (≈0.12).

- Swedish is the next closest (0.039), confirming the coherence of the North Germanic subgroup.
- English and German fall at comparable distances (0.119 and 0.124), placing both in a distinctly more distant tier relative to Danish.

8.4 Interpretation

WALS-based cosine distance is a clean, reproducible proxy for structural similarity, but it does not capture the full picture. Pretraining data volume and tokenizer overlap also shape how well a multilingual encoder represents a given language — and those factors are not encoded in WALS vectors. What the distances do support is an unambiguous ordering: NO $<$ SV $<$ EN $\approx$ DE. That ordering is directionally consistent with the NO$\rightarrow$DA $>$ EN$\rightarrow$DA pattern observed in zero-shot experiments, even when individual transfer gaps fall short of statistical significance.

9 Results

9.1 Multi-seed zero-shot transfer results

Table 4 reports the core comparison — Norwegian vs. English zero-shot transfer — across seeds 42, 1 and 2, with per-system means, standard deviations and paired t-test p-values.

Three observations stand out. First, seed 42 made Norwegian XLM-R look clearly better on dev (86.49 vs. 85.52) — until you average across seeds, at which point English XLM-R pulls slightly ahead (85.87 vs. 85.45). That reversal encapsulates the cautionary message: single-seed development comparisons can mislead. Second, on the held-out test set Norwegian leads for both encoders: +1.86 F1 for XLM-R ($p = 0.10$) and +3.36 F1 for mBERT ($p = 0.046$). Third, the mBERT result is the more reliable of the two — a larger margin, lower variance, and conventional significance despite the limited number of usable seed pairs.

System	S42	S1	S2	Mean	Std	p
Development set (Danish dev)						
NO→DA XLM-R	86.49	84.98	84.90	85.45	0.89	0.632 ns
EN→DA XLM-R	85.52	86.57	85.52	85.87	0.60	
NO→DA mBERT	83.38	83.40	84.52	83.77	0.65	0.650 ns
EN→DA mBERT	—	82.23	83.68	82.95	1.02	
Test set (Danish test)						
NO→DA XLM-R	82.46	83.47	83.41	83.11	0.56	0.100 ns
EN→DA XLM-R	81.86	81.15	80.75	81.25	0.56	
NO→DA mBERT	82.88	84.06	80.96	82.63	1.56	**0.046***
EN→DA mBERT	—	78.98	79.56	79.27	0.41	

Table 4: Multi-seed strict F1 (%) for zero-shot transfer to Danish. "—" marks the EN mBERT seed-42 dev run which is a known artifact (Section 10) and is excluded. p-values from paired t-tests (NO vs EN within encoder). * $p < 0.05$.

Setup	Strict F1	Unlabeled F1	Loose F1
Source-language baselines			
Norwegian XLM-R	92.0	95.9	92.8
Norwegian BERT	91.2	95.1	92.1
Danish BERT	89.0	93.0	92.2
Danish XLM-R	88.6	92.6	91.8
English XLM-R	81.7	86.3	85.7
Zero-shot transfer to Danish			
NO→DA XLM-R	86.5	90.3	90.7
EN→DA XLM-R	85.5	90.7	88.5
NO→DA mBERT	83.4	88.2	86.6
Fine-tuned on Danish			
EN→DA mBERT + DA FT	89.4	93.4	91.7
NO→DA XLM-R + DA FT	88.4	92.7	91.5
NO→DA mBERT + DA FT	87.6	92.1	90.5
EN→DA XLM-R + DA FT	87.0	90.8	90.4
Danish data efficiency			
Danish XLM-R 50%	85.5	90.8	90.0
Danish XLM-R 25%	82.8	88.8	88.3
Danish XLM-R 10%	70.0	77.2	81.7

Table 5: Development-set span F1 scores (%). Strict = exact span and label; Unlabeled = exact span, any label; Loose = partial overlap with matching label.

9.2 Development set — span F1 metrics

Table 5 presents strict, unlabeled and loose span F1 for the single-seed (seed 42) runs. The anomalous EN→DA mBERT zero-shot run (strict ≈ 0.3) is excluded from comparative summaries and examined in Section 10.

Unlabeled and loose scores exceed strict by 3–5 points across most systems — most errors are near-misses in boundary or label, not completely missed spans. The largest strict-to-loose gap appears at 10% Danish data (70.0 vs. 81.7), suggesting that with very limited supervision the model often identifies the right span but lacks the signal to assign the correct label reliably.

9.3 Test set results

Table 6 reports strict, unlabeled and loose F1 on held-out test files. Danish-target experiments are evaluated on the Danish DDT test split; Norwegian

Setup	Strict F1	Unlabeled F1	Loose F1
Source-language baselines			
Norwegian BERT	92.2	94.6	93.0
Norwegian XLM-R	91.4	94.4	92.9
Danish BERT	85.3	90.2	86.6
Danish XLM-R	84.9	89.9	87.2
Zero-shot transfer to Danish			
EN→DA XLM-R	81.9	87.7	83.1
NO→DA XLM-R	82.5	88.7	84.0
NO→DA mBERT	82.9	88.8	84.4
Fine-tuned on Danish			
EN→DA mBERT + DA FT	85.5	89.5	87.0
NO→DA XLM-R + DA FT	86.5	90.4	87.7
NO→DA mBERT + DA FT	86.6	90.2	88.1
EN→DA XLM-R + DA FT	84.7	88.2	87.1
Danish data efficiency			
Danish XLM-R 50%	83.6	88.9	85.6
Danish XLM-R 25%	84.1	88.6	86.7
Danish XLM-R 10%	72.8	78.2	84.4

Table 6: Test-set span F1 scores (%). Danish transfer experiments are evaluated on the Danish DDT test split; Norwegian baselines on the NorNE test split.

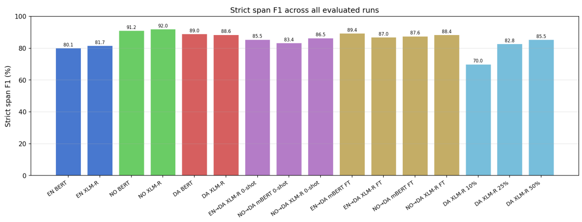

Figure 1: Strict span F1 (%) for all evaluated runs on the development set. Colors group run types: blue = English baselines, green = Norwegian baselines, red = Danish baselines, purple = zero-shot transfer, yellow = fine-tuned transfer, teal = Danish data efficiency.

source baselines on the NorNE test split. English monolingual test results are omitted because the English test split is masked.

Most Danish-target test scores fall 3–5 points below their dev counterparts, a normal generalization gap. Norwegian source baselines remain high on NorNE (92.2 BERT, 91.4 XLM-R). After Danish fine-tuning, Norwegian and English transfer models converge in a narrow band (approximately 84.7–86.6), while zero-shot systems still trail by 2–4 points.

9.4 Figures

Figure 1 shows the distribution of development strict F1 across all runs. Norwegian source baselines rank highest; Danish fine-tuning recovers most of the gap to Danish monolingual baselines.

Figure 2 plots the Danish XLM-R learning curve alongside zero-shot reference lines. The curve crosses the English zero-shot baseline between 25% and 50% Danish data and approaches the Norwegian zero-shot line shortly beyond 50%, indicating that moderate annotation recovers most of the zero-shot advantage quickly.

Figure 3 compares strict F1 from saved predictions with training-time evaluation F1. All runs except the known artifact cluster tightly near the diagonal.

Figure 4 provides a compact visual summary highlighting the Norwegian advantage and the EN→DA zero-shot artifact.

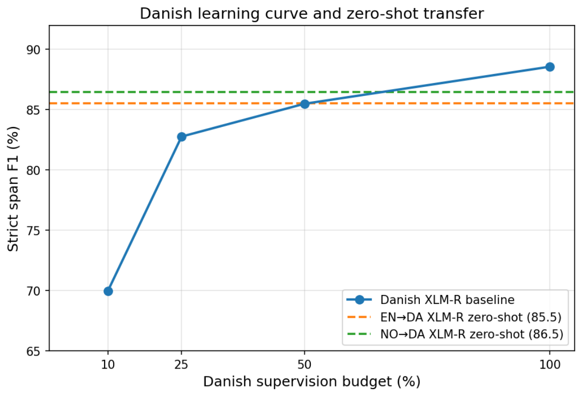

Figure 2: Danish XLM-R learning curve (solid) and zero-shot reference lines (dashed). The learning curve reaches the zero-shot baselines between 25% and 50% Danish data.

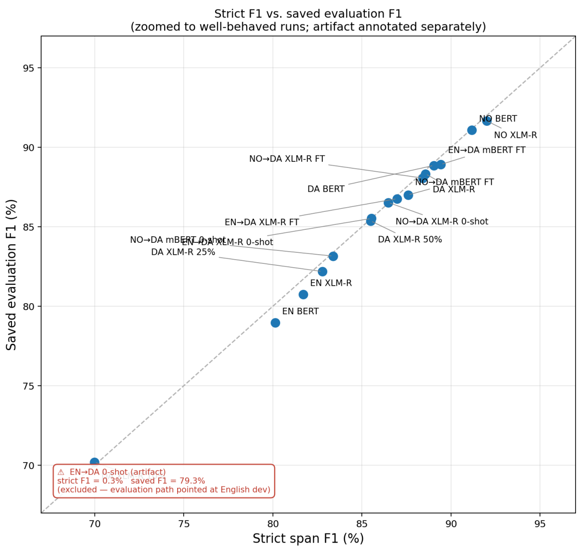

Figure 3: Strict span F1 vs. saved evaluation F1 (zoomed to well-behaved runs). The EN→DA zero-shot artifact — strict F1 ≈0.3%, saved F1 = 79.3% — is excluded from the axis range and annotated in the corner.

9.5 Per-entity-type F1

Table 7 shows per-type F1 for key systems on the Danish development set.

ORG is the most variable category across sys-

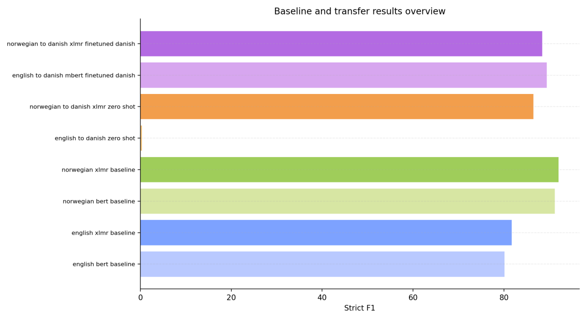

Figure 4: Compact overview of source-language and transfer run results.

System	PER	LOC	ORG
Danish BERT baseline	91.6	91.3	82.4
Danish XLM-R baseline	93.7	89.3	79.6
EN→DA XLM-R zero-shot	89.8	86.5	77.3
NO→DA XLM-R zero-shot	91.0	90.0	74.9
EN→DA mBERT + DA FT	91.0	92.7	83.5
NO→DA mBERT + DA FT	89.6	90.5	81.7
NO→DA XLM-R + DA FT	93.1	90.6	78.8

Table 7: Per-entity-type strict F1 (%) on Danish dev. MISC is excluded: those annotations appear only in a secondary label column not read by the training and evaluation code, so MISC F1 is 0.0 for all systems by construction (see Section 10).

tems, swinging from 74.9 to 83.5 — a spread that far exceeds what either PER or LOC shows. The EN→DA mBERT + DA FT model records the highest ORG score (83.5), which contributes to its strong overall ranking in Table 5. LOC shows a notable zero-shot advantage for Norwegian XLM-R (90.0 vs. 86.5 for English), plausibly because place names share more lexical surface form between Norwegian and Danish. PER is the most stable category across all systems (89.6–93.7), consistent with the broader finding that person names transfer more reliably across languages.

9.6 Distance and gain

10 Logs and Output Analysis

10.1 The anomalous EN→DA zero-shot prediction file

One English-to-Danish zero-shot BERT run (`english_to_danish_zero_shot`) produced a dev prediction file with 2,001 sentences, while the Danish dev gold contains only 564. Evaluated against the Danish gold, this gives strict F1 of approximately 0.3% — even though the training log reported 79.3. The most plausible explanation is a stale or incorrect evaluation path:

Source	WALS Distance	Zero-shot F1 (dev)	Test F1
Norwegian XLM-R	0.021	86.5	82.5
Norwegian mBERT	0.021	83.4	82.9
English XLM-R	0.119	85.5	81.9

Table 8: WALS syntactic distance (lang2vec) versus zero-shot transfer performance. Norwegian is 5.7× closer to Danish than English. The transfer advantage is directionally consistent but not statistically significant at the dev level (Table **??**).

the trainer appears to have pointed its evaluation call at the English development split, saved that score under the Danish dev label, and the resulting predictions simply do not align with the correct Danish gold file.

Blast-radius check. To determine whether the bug propagated, we compared sentence counts between saved prediction files and their intended gold splits. Only `english_to_danish_zero_shot` shows a mismatch on dev (2,001 vs. 564); the remaining 16 experiments produce files with the expected counts. We also checked the gap between strict F1 from saved predictions and the eval F1 logged during training: aside from this artifact, all gaps fall below approximately 2 F1 points, within the range expected from minor tokenisation and boundary-handling differences. The bug appears isolated to this single run and did not propagate to fine-tuned models initialized from its checkpoint — the checkpoint itself represents a valid English NER model, and subsequent Danish fine-tuning used correct Danish supervision.

On the test split the same experiment shows no sentence-count mismatch and records a test strict F1 of 78.8, consistent with the stored metric. This further supports the diagnosis that only the dev-evaluation path was mis-specified. We exclude the erroneous dev result from comparative analyses but report the test score for completeness.

10.2 Why MISC F1 is 0.0 on Danish

The per-entity-type tables show MISC F1 of 0.0 for every system. This is a data-format issue rather than a modelling failure. The Danish Universal NER file stores primary DDT-style labels (PER, LOC, ORG, O) in the main annotation column, while MISC labels appear in a secondary column (e.g. `B-MISC#O`). Both the training code and `span_f1.py` read only the primary column, so MISC annotations are never seen during training and never appear in the evaluated label stream.

The Danish dev and test files contain 32 and 44 MISC annotations respectively, but these reside exclusively in the secondary column.

10.3 Metric consistency for well-behaved runs

Setting aside the artifact, the remaining 16 runs show close agreement between strict span F1 from saved prediction files and the eval F1 logged during training (see Figure 3). The largest observed discrepancy is roughly 1.2 F1 points (Danish XLM-R 25%), which falls within the range expected from minor differences in tokenisation and boundary handling between evaluation paths. This cross-check confirms that the pipeline produced reproducible and internally consistent scores for all properly configured runs.

11 Discussion

11.1 What multi-seed replication revealed

A single seed can give a misleading picture. Seed 42 made Norwegian XLM-R look clearly better on the development set (86.49 vs. 85.52), but once averaged across three seeds, English XLM-R pulls slightly ahead on dev (85.87 vs. 85.45). That reversal is the cautionary finding in one line: cross-lingual source-language comparisons based on a single initialization are unreliable.

The held-out test set tells a cleaner story. Norwegian leads for both encoders — +1.86 F1 for XLM-R ($p = 0.10$, borderline) and +3.36 F1 for mBERT ($p = 0.046$, significant). The mBERT result is the more convincing of the two: a larger margin, lower variance, and conventional significance despite the limited number of usable seed pairs. The XLM-R effect is directionally consistent, but additional seeds would be needed to settle its significance. The takeaway is not that development evaluation is useless; it is that held-out test results are essential for source-language claims, and replication should not be optional.

11.2 Inconvenient findings

Two results complicate a simple distance-based narrative and are worth confronting directly.

English mBERT wins after fine-tuning. The best transfer-plus-fine-tune system is English mBERT fine-tuned on Danish (89.4 dev, 85.5 test), outperforming all Norwegian transfer variants. This looks like a contradiction of the typological-distance hypothesis, until you recall that English provides roughly 20,000 training sentences against approximately 4,000 for Norwegian. A larger source initialization can outweigh typological proximity once target-language supervision is available to steer the model. The distance hypothesis is most informative in the pure zero-shot setting, where that data-volume advantage cannot compensate.

Danish XLM-R underperforms Danish BERT in monolingual training. Danish XLM-R trails Danish BERT slightly (88.6 vs. 89.0 dev; 84.9 vs. 85.3 test). XLM-R's broader multilingual capacity allocation may yield slightly less Danish-specific representations compared with mBERT in this fully supervised setting — or the gap may simply fall within run-to-run noise and disappear under multi-seed replication. Either interpretation cautions against assuming XLM-R is uniformly superior.

11.3 Encoder choice

XLM-R tends to outperform mBERT in source-language baselines and zero-shot scenarios (e.g. 92.0 vs. 91.2 for Norwegian baselines), reflecting its larger pretraining corpus. That advantage is most apparent without target-language supervision. After Danish fine-tuning, the gap narrows or sometimes reverses — models with more concentrated vocabulary allocation, such as mBERT, appear to catch up once target labels are available to specialize the representations.

11.4 Data efficiency

The Danish learning curve (Figure 2) shows that zero-shot transfer provides a reasonable starting point, and that moderate annotation recovers most of the remaining gap to full supervision quickly. Reaching approximately 25–50% of the Danish training set is enough to match or surpass the zero-shot baselines. In practice, one could start with a Norwegian zero-shot model and label a fraction of a target dataset to reach competitive performance — a practical trade-off when annotation resources are limited.

11.5 Limitations

Several limitations bound what can be concluded. First, multi-seed replication covers only three seeds for a subset of comparisons, leaving some effect-size estimates uncertain. Second, only English and Norwegian are used as source languages; experiments with Swedish or German would test whether a typological distance gradient corresponds to a measurable F1 gradient. Third, results come from a single NER benchmark family, so generalization to other domains or annotation conventions is untested. Fourth, no per-source hyperparameter tuning was performed, which may have disadvantaged certain configurations.

12 Conclusion

We ran a multi-seed study of cross-lingual NER transfer to Danish, comparing English and Norwegian as source languages under mBERT and XLM-R. Systems are evaluated with strict, unlabeled, and loose span F1 on both development and held-out test sets, and the English–Norwegian contrast is assessed with paired resampling tests.

Main finding. In the zero-shot setting, Norwegian is the better source on the Danish test set for both encoders: a statistically significant gain for mBERT ($\Delta = +3.36$, $p = 0.046$) and a smaller, directionally consistent effect for XLM-R ($\Delta = +1.86$, $p = 0.10$). One development result — seed 42, which had favored Norwegian XLM-R — turned out to be an outlier, confirming that held-out test replication is necessary before drawing source-language conclusions.

Why Norwegian appears stronger. Norwegian is roughly 5.7× closer to Danish than English in WALS syntactic-feature space (cosine distance 0.021 vs. 0.119). This structural proximity aligns with the zero-shot advantage, though data volume and encoder-level factors also play a role.

When English outperforms. Once Danish fine-tuning is available, the best result comes from English mBERT (89.4 dev, 85.5 test). English provides roughly five times as many source-language training sentences as Norwegian, and that initialization advantage can outweigh typological proximity when target supervision is available to steer the model. The distance hypothesis is therefore most informative in the pure zero-shot setting.

Entity-type and metric notes. ORG is the most variable category, spanning nearly nine F1 points across systems (74.9–83.5 on Danish dev), while PER is the most stable (89.6–93.7). MISC F1 is 0.0 across all systems because those annotations reside in a secondary label column not read by the training and evaluation code. Unlabeled and loose F1 consistently exceed strict by 3–5 points, with the largest gaps under low-resource Danish training where the model detects spans but mislabels them.

Practical recommendation. For zero-shot Danish NER, Norwegian is the linguistically motivated and empirically supported source choice. If any Danish annotation budget is available, fine-tuning an English mBERT checkpoint on Danish is a strong practical baseline. Full Danish supervision remains the performance ceiling.

Next steps. The most valuable extensions would be: (1) additional seeds for XLM-R to resolve the marginal p-value; and (2) transfer experiments from Swedish — and possibly German — to test whether a typological distance gradient produces a corresponding F1 gradient across more than two source languages.

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